

## **Course Name: Java Programming**

**Course Code:**      **Internal Marks: 25**      **External Marks: 75**      **Credit: 4**

### **UNIT-I**

**JAVA DATA TYPES AND OPERATORS:-** Genesis of Java: Creation of Java – why java is important to internet – The java Buzz words – An overview of Java Object Oriented Programming. Data types: Simple types – Integers – Floating point types – characters – Booleans – A closer Look at Literals – Variables – Type conversion and casting – Automatic type promotion in Expressions – Strings. Arrays: One Dimensional Array – Multi Dimensional Array. Operator: Arithmetic Operators – Bitwise operators – Relational operators – Boolean Logical operators – Assignment operators – Conditional operators–Operator Precedence.

### **UNIT-II**

**INTRODUCING CLASSES, METHODS AND INHERITANCE:-** Class Fundamentals – Declaring objects – Assigning object Reference variables – Introducing Methods – Constructors – Garbage collection – Finalize () Method – Stack class. A Closer Look at Methods and classes: Overloading Methods – Using object as parameters – Argument passing – Returning objects – Recursion – Introducing Access control – understanding static – Introducing final – Nested and Inner classes – String class – Using command line arguments. Inheritance Basics – Using super – creating Multilevel Hierarchy – Method overriding – Dynamic Method Dispatch – Using Abstract class – Using final with inheritance – The object class.

### **UNIT-III**

**PACKAGES,INTERFACES, EXCEPTION HANDLING AND MULTITHREADING:-** Packages – Access Protection – Importing packages – Interfaces. Exception Handling Introduction – Exception Types – Uncaught Exceptions – Using try and catch – Multiple catch clauses – Nested try statements – throw- throws- finally – Java's Built – in Exception – creating your own Exception subclasses. Multithreaded Programming: Java Thread Model – Main Thread – Creating a Thread - Creating Multiple Threads–Using is Alive () and join () – Thread priorities – Synchronization – Inter thread Communication – Suspending Resuming: and stopping Threads – Using Multithreading.

## **UNIT-IV**

**APPLETS AND EVENT HANDLING:-** I/O, Applets and other topics: I/O Basics Reading console Input – writing console output – The Print Writer class – Reading and Writing Files. The Applet class: Applet Basics – Applet Architecture – Applet Skeleton – Applet Display method – Requesting Repainting – HTML APPLET tag- Passing Parameters to Applet – Audio Clip Interface. Event Handling Mechanisms – Delegation Event Model – Event classes (The Action Event Item Event, Key Event, Mouse Event) – Sources of Events – Event Listener Interfaces (Action Listener, Item Listener, Key Listener, Mouse Listener) – Adapter Classes.

## **UNIT-V**

**INTRODUCING AWT AND AWT CONTROLS:-** AWT Classes – Window fundamentals – working with Frame Windows - working with Graphic Using AWT controls: Controls fundamentals – Labels – using Buttons – Applying check Boxes – Check Box group – Choice controls – Using a Text field – Using a Text Area – Understanding Layout Managers (Flow Layout only) – Menu Bars and Menus.